Back-of-the-Envelope Estimation

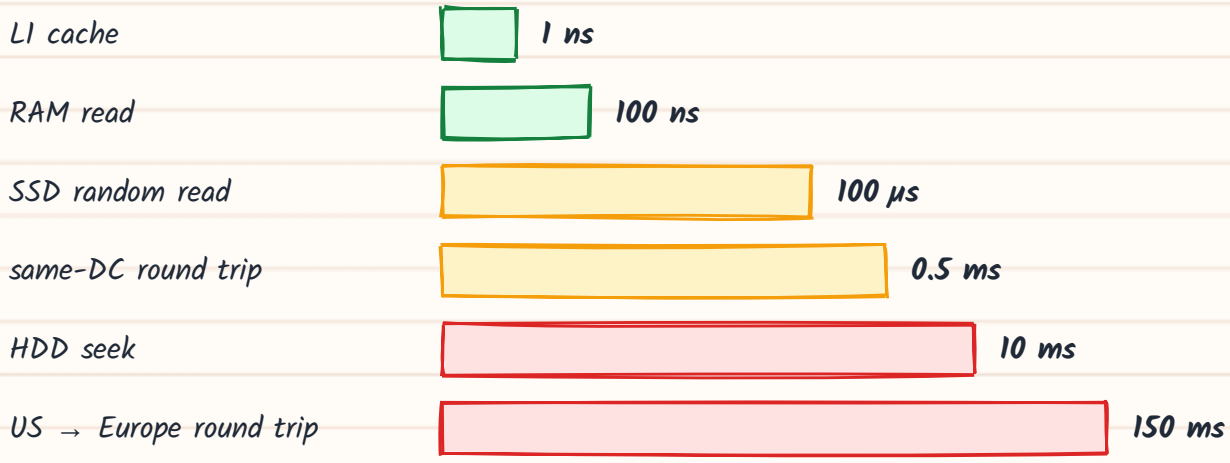
Lesson 2 · latency numbers, QPS, storage & cache size

In this lesson → ① Why estimate ② Latency numbers ③ Powers of two ④ QPS ⑤ Reads vs writes ⑥ Storage ⑦ Bandwidth & cache ⑧ Interview tips ⑨ Interview Q&A
Running project: SnapLink, our photo-sharing app that grows from 1 server to 100M users.

1 Why estimate?

> **Back-of-the-envelope estimation** = quick math for the order of magnitude of traffic, storage, bandwidth and memory, so each design choice has a reason.

2 Latency numbers (classic, order of magnitude)



Operation	Time	Feels like
L1 cache reference	~1 ns	instant
Main memory reference	~100 ns	very fast
SSD random read	~100 µs	1,000× RAM
Round trip in same data center	~0.5 ms	
HDD disk seek	~10 ms	100,000× RAM
US ↔ Europe round trip	~150 ms	noticeable

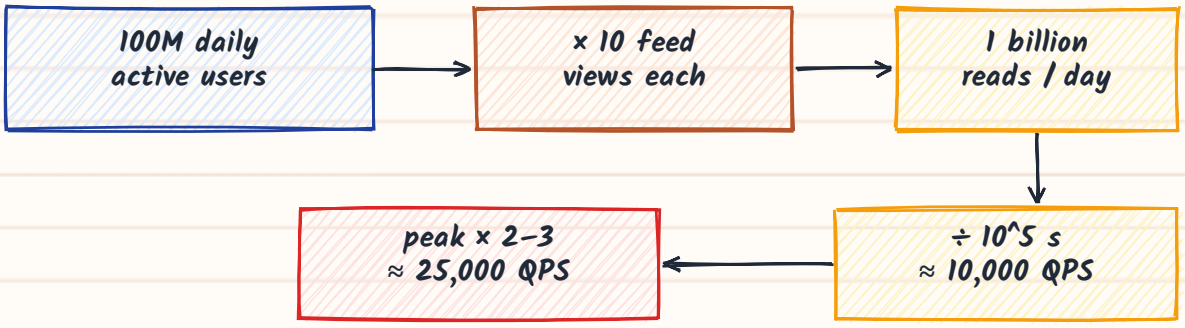
3 Powers of two & units

2^10	≈ 1 thousand	KB
2^20	≈ 1 million	MB
2^30	≈ 1 billion	GB
2^40	≈ 1 trillion	TB
2^50	≈ 1 quadrillion	PB

1 day ≈ 86,400 s ≈ 10^5 s

Handy number	Value
Seconds per day	86,400 ≈ 10^5
Seconds per month	≈ 2.5 × 10^6
Seconds per year	≈ 3 × 10^7
1 million requests/day	≈ 12 QPS

4 Estimating QPS



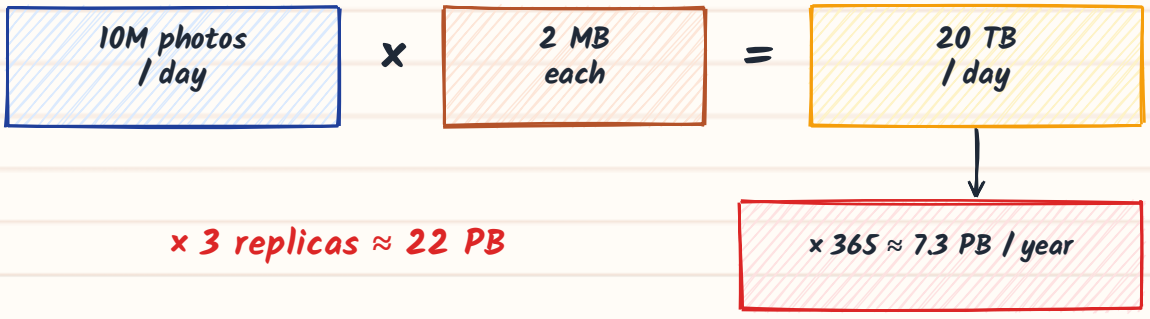
Formula
Average QPS = DAU × requests per user per day ÷ 10^5 · **Peak** ≈ 2–3 × average

5 Reads vs writes

SnapLink	Per day	Per second
Feed reads	1 billion	≈ 10,000 (peak 25k)
Photo uploads	10 million	≈ 100

System type	Design choices
Read-heavy	cache, read replicas, CDN, denormalised reads
Write-heavy	sharding, queues/batching, LSM-tree stores

6 Estimating storage



7 Bandwidth & cache memory

Estimate	How	SnapLink
Ingress bandwidth	bytes/day ÷ 10^5	20 TB ÷ 10^5 ≈ 200 MB/s
Egress bandwidth	reads × object size	much larger → CDN
Cache memory	20% of daily read data (80/20)	decides # cache nodes
Servers	peak QPS ÷ QPS per server	e.g. 25k ÷ 2.5k ≈ 10

8 Interview tips

- ✓ Round to powers of 10
- ✓ State assumptions out loud
- ✓ Write numbers on the board
- ✓ Always end with a decision ("so we need...")

Example conclusion
"~25k peak QPS, 7 PB/year of photos → ~10 app servers behind an LB, a cache cluster, photos in object storage + CDN."

Interview Corner

Q. How do you estimate QPS?

A. DAU × requests per user ÷ ~10^5 seconds; multiply by 2–3 for peak.

Q. How much slower is a disk seek than memory?

A. ~10 ms vs ~100 ns: about 100,000×.

Q. Why estimate storage?

A. To pick the storage type, plan sharding and replication, and predict cost.

Q. What is the 80/20 rule for caching?

A. ~20% of data serves ~80% of requests; size the cache for that hot set.

Q. How precise must estimates be?

A. Order of magnitude is enough; the reasoning and assumptions matter more.

⚡ Revise in 60 seconds

- → RAM **100 ns** · SSD **100 µs**
- → DC trip **0.5 ms** · seek **10 ms**
- → Cross-continent **150 ms**
- → 1 day ≈ **10^5 s**
- → QPS = DAU × req ÷ 10^5
- → Peak ≈ 2–3×
- → Storage = items × size × days
- → Cache = hot 20%

Next lesson → SD-03
Scalability: vertical vs horizontal, stateless servers.
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